

Algebra General Exam - August 2025

Problem 1

Let $\mathbb{F}_q$ be a finite field with q elements.

(a) Find the orders of the general linear group $\mathrm{GL}_2(\mathbb{F}_q)$ and special linear group $\mathrm{SL}_2(\mathbb{F}_q)$: (4 points)

$$\mathrm{GL}_2(\mathbb{F}_q) := \{A \in \mathrm{M}_2(\mathbb{F}_q) \mid \det A \neq 0\}, \quad \mathrm{SL}_2(\mathbb{F}_q) := \{A \in \mathrm{M}_2(\mathbb{F}_q) \mid \det A = 1\}.$$

(b) Show that $\mathrm{SL}_2(\mathbb{F}_q)$ contains elements of orders $q - 1$ and $q + 1$. (*Hint.* To find an element of order $q + 1$, use the extension $\mathbb{F}_{q^2} / \mathbb{F}_q$.) (6 points)

Problem 2

As usual, S_6 and A_6 will denote the symmetric and alternating groups on 6 elements.

(a) Let $\sigma = (123)(456) \in A_6$. Find the sizes of the conjugacy classes $C_{S_6}(\sigma)$ and $C_{A_6}(\sigma)$. (5 points)

(b) Let $\tau = (12345) \in A_6$. Find the sizes of the conjugacy classes $C_{S_6}(\tau)$ and $C_{A_6}(\tau)$. (5 points)

Problem 3

The symmetric group $G = S_4$ has the following character table, where for each conjugacy class C_i we indicate a representative and its size c_i .

Character table of the symmetric group S_4

C_i	(1)	(12)	(12)(34)	(123)	(1234)
c_i	1	6	3	8	6
χ_1	1	1	1	1	1
χ_2	1	-1	1	1	-1
χ_3	2	0	2	-1	0
χ_4	3	1	-1	0	-1

C_i	(1)	(12)	(12)(34)	(123)	(1234)
χ_5	3	-1	-1	0	1

(a) Use the table to determine the decomposition of the representation $\rho_4 \otimes \rho_5$ into the direct sum of irreducible representations, where ρ_i denotes the irreducible representation of G with character χ_i . (5 points)

(b) Let $G = S_n$ with $n \geq 4$. Let ρ be the standard $(n - 1)$ -dimensional representation of G (“permutation representation minus the trivial representation”), and let σ be the sign representation (in other words, the 1-dimensional representation whose kernel is the alternating group A_n). Show that the representations ρ and $\rho \otimes \sigma$ are not equivalent. (5 points)

Problem 4

Let $M = \mathbb{Z}^n$ be a free $\mathbb{Z}$ -module of rank $n \geq 2$. We recall that the group $\text{SL}_n(\mathbb{Z}) = \{A \in \text{M}_n(\mathbb{Z}) \mid \det A = 1\}$ acts on M by automorphisms through matrix multiplication. Set

$$X = \{(a_1, \dots, a_n) \in M \setminus \{0\} \mid \text{g.c.d.}(a_1, \dots, a_n) = 1\}.$$

(a) Show that for any $w \in X$ there exists a basis $v_1, \dots, v_n$ of M such that $v_1 = w$. (4 points)

(b) Let $w = (6, 10, 15)$. Find a basis v_1, v_2, v_3 of $M = \mathbb{Z}^3$ such that $v_1 = w$. (3 points)

(c) Use part (a) to show that $\text{SL}_n(\mathbb{Z})$ acts on X transitively. (3 points)

Problem 5

Let F be a field of characteristic 5. Classify up to conjugacy all (5×5) -matrices X over F satisfying $X^5 = I_5$. (10 points)

Problem 6

Let $R = \mathbb{C}[x, y]$ be the ring of complex polynomials in two variables. For an element $a \in R$, we let (a) denote the principal ideal generated by a . Prove the following properties of the ring R .

(a) While the g.c.d. of x and y is 1, show that the natural homomorphism

$$R \rightarrow R/(x) \times R/(y)$$

is not surjective. (4 points)

(b) Prove that no principal ideal of R is maximal. (6 points)

Problem 7

Let F be a field, and let $f(x), g(x) \in F[x]$ be nonconstant polynomials. Then the quotients $F[x]/(f(x))$ and $F[x]/(g(x))$ can be considered as $F[x]$ -modules/algebras as well as F -vector spaces/algebras.

(a) Use the universal property of tensor product to show that there exists a surjective homomorphism of F -algebras (4 points):

$$\gamma : \frac{F[x]}{(f(x))} \otimes_F \frac{F[x]}{(g(x))} \rightarrow \frac{F[x]}{(f(x))} \otimes_{F[x]} \frac{F[x]}{(g(x))}.$$

(b) Show that γ is an isomorphism if and only if f and g are proportional linear polynomials. (6 points)

Problem 8

Let E/K be a cyclic Galois extension of degree n with Galois group $G = \langle \sigma \rangle$. Show that there exists $a \in E$ such that $a, \sigma(a), \dots, \sigma^{n-1}(a)$ is a K -basis of E . (Hint. Consider σ as a K -linear automorphism of the K -vector space E . Determine the minimal polynomial of σ , and use this information to conclude that E has a “cyclic” basis for σ .) (10 points)